

**1 0**

Read the following passage.

Tropical rainforests are among the most biologically diverse ecosystems in the world. Their varied plant communities support a high biodiversity of animal species. Tropical rainforests form in parts of the world with high annual rainfall and high mean temperatures providing moist, warm conditions. Nutrient recycling occurs rapidly on the ground of these rainforests. The recycling of nitrogen-containing compounds from dead organic matter sustains the growth of plants.

5

Each year, large areas of rainforests are lost due to fires. Natural regeneration of rainforests occurs slowly by secondary succession. Secondary succession is similar to primary succession but occurs in areas previously inhabited. There is conflict between the needs of humans and conservation of rainforests. Deforestation by humans is removing large areas of rainforests. Farming practices have prevented the natural regeneration of rainforests by secondary succession.

10

Use the information in the passage and your own knowledge to answer the following questions.

**1 0****. 1**

In tropical rainforests, there is a high biodiversity of animal species (lines 1–3).

Explain why.

**[3 marks]**

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1 0 . 2

Nutrient recycling occurs rapidly on the ground of tropical rainforests (lines 4–5).

Explain how **two** abiotic factors in these rainforests enable the rapid rate of recycling.

[2 marks]

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1 0 . 3

The recycling of nitrogen-containing compounds from dead organic matter sustains the growth of plants (lines 5–7).

Describe the roles of bacteria in this recycling.

[4 marks]

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Question 10 continues on the next page

Turn over ►

**1 0 . 4**

Natural regeneration of rainforests occurs by secondary succession (lines 8–10).

Describe the process of secondary succession.

**[4 marks]**

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**1 0 . 5**

Farming practices have prevented the natural regeneration of rainforests by secondary succession (lines 12–14).

Suggest **two** farming practices that prevent the natural regeneration of rainforests by secondary succession.**[2 marks]**

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**1****END OF QUESTIONS**

[illegible]

[illegible]

| Question number | Additional page, if required.<br>Write the question numbers in the left-hand margin.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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